

PRINCIPAL AI ARCHITECTURE CASE STUDY

MULTI-TENANT AGENTIC AI PLATFORM

Enterprise AI Operating System

ARCHITECTURAL MANDATE

Build reusable AI platform primitives instead of repeatedly solving tenant isolation, orchestration, knowledge access, tool authority, model integration, evaluation and operations.

DESIGN THESIS

Probabilistic AI belongs inside deterministic systems. Model decisions can influence workflow direction, but execution authority remains bounded by policy, state and explicit contracts.

8,500+

Concurrent AI sessions — cover evidence set

99.97%

Platform availability — cover evidence set

ANONYMIZED • ENTERPRISE AI ARCHITECTURE CASE STUDY

Unified edition consolidating the supplied reference PDF into one professionally typeset, text-only architecture artifact. All architecture diagrams have intentionally been removed to eliminate truncation and right-edge rendering issues.

Evidence integrity note: the supplied reference contains multiple evidence sets in different sections — notably 8,500+ vs 1,000+ concurrent sessions and 99.97% vs 99.94% availability. This edition preserves the source values rather than silently reconciling them. Validate the intended final figures before external circulation.

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A single architecture artifact — no embedded diagrams.

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ARCHITECTURE SCOPE

Enterprise AI control plane, agent runtime, retrieval, model gateway, memory, governance, observability, security, resilience, cloud infrastructure and production operations.

01

EXECUTIVE ARCHITECTURE OVERVIEW

Reusable enterprise AI control plane + execution substrate

The platform is designed as a reusable enterprise AI control plane and execution substrate. The central architectural separation is between governance/control concerns and the runtime data plane: registries, policy, risk, model status and governance decisions constrain what the runtime can execute.

Architecture layer	Primary responsibility	Hard architectural question
Experience / Edge	API, event and conversational entry; request context	Who is calling, for which tenant, and under what purpose?
Control Plane	Agent/tool/model registries, policy, governance	Who is allowed to authorize AI behaviour?
Agent Data Plane	Durable workflow execution, state, action validation	Where does probabilistic behaviour meet deterministic control?
Knowledge / Context	Graph, vector, memory, ACL, provenance, context budget	What evidence is admissible and how much context is safe?
Interoperability	MCP tool boundary and A2A delegation contracts	How is capability authority explicitly bounded?
Inference	Routing, caching, serving, batching, fallback	How are quality, latency and economics jointly optimized?
Operations	Telemetry, SLOs, evaluation, audit, FinOps	How is production AI behaviour measurable and reconstructable?

Architecture is treated as a system of explicit boundaries: identity, tenant context, authority, workflow state, evidence, inference, execution, observability and auditability.

ARCHITECTURE MANDATE & DESIGN THESIS

The architectural objective is not to build a single agent. It is to establish reusable platform primitives so multiple enterprise AI capabilities can operate without each application independently solving tenant isolation, orchestration, knowledge access, model integration, tool authority, evaluation and operations.

CONTROL Policy and governance precede execution	RUNTIME Durable, stateful workflow execution	EVIDENCE Retrieval + provenance + audit
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PROBABILISTIC AI INSIDE
DETERMINISTIC SYSTEMS

The model can propose, rank, summarize, retrieve and reason over enterprise context. Deterministic systems retain authority over identity, policy, state transitions, external side effects, evidence and recovery.

Production posture

Multi-tenant isolation • durable execution • governed tools • evidence-aware retrieval • model routing • evaluation gates • zero-trust security • SRE • FinOps

The platform standard is not whether an agent can complete a task in a demo. It is whether the system can explain who had authority, what evidence was used, what decision was made, what action was executed, what failed, how it recovered and what it cost.

FLAGSHIP TECHNICAL CASE STUDY

Enterprise Agentic AI Orchestration Platform — anonymized

Problem statement

The architecture was application-centric: individual AI workloads repeatedly solved tenant isolation, orchestration, knowledge access, model integration, tool authority, evaluation and operational controls. This created duplicated platform concerns and made consistent governance, failure containment, observability and economics harder to enforce.

Objective

Establish reusable enterprise AI platform primitives that allow autonomous and stateful workloads to scale while preserving deterministic authority boundaries, durable execution semantics, evidence-aware retrieval, provider abstraction, governed tool access, measurable AI quality and operational accountability.

Architectural scope

Plane	Core responsibility	Representative mechanisms
Control plane	Tenant, agent, tool and policy intent	Tenant configuration, capability registration, governance metadata
Data plane	Runtime execution	AKS, LangGraph, distributed services, asynchronous processing
Knowledge plane	Enterprise knowledge and context	Neo4j GraphRAG, vector retrieval, context assembly
Interoperability plane	Controlled external and agent capabilities	MCP and A2A with identity and policy
Inference plane	Model abstraction and economics	Azure AI Foundry, approved providers, routing, semantic caching
Trust plane	Privacy, safety and evaluation	Presidio, policy controls, RAGAS, DeepEval, Giskard, PyRIT
Operations plane	Production evidence	OpenTelemetry, reliability signals, cost telemetry

Core architecture decisions

- Execution substrate: controlled graph-based workflow orchestration with durable execution state and explicit state transitions.
- Knowledge: GraphRAG where relationship structure matters; vector retrieval for semantic matching; provenance-aware context assembly and token budgeting.
- Authority: governed MCP tool boundary; capability contracts for A2A delegation; enterprise credentials remain outside the model.
- Inference: model abstraction and routing across quality, latency, capability and economics; authorization-aware semantic caching.
- Eventing: durable asynchronous coordination, tenant/workflow partitioning, idempotency, replay and DLQ handling.
- Quality: repeatable retrieval, model, agent and safety evaluation with shadow/canary release gates.
- Operations: OpenTelemetry-based tracing, AI quality telemetry, SLOs, cost telemetry, automated remediation and immutable audit evidence.
- Security: zero-trust runtime boundaries, policy-as-code, short-lived credentials, least privilege and adversarial regression controls.

ARCHITECTURE LIFECYCLE

Business intent → requirements → constraints → architecture → implementation → validation → production feedback

The working model is deliberately front-loaded with architecture reasoning. A request is not translated directly into an agent, RAG pipeline or model choice. It is decomposed into business capability, functional behavior, NFRs, governing constraints, AI suitability, ownership boundaries and failure semantics before technology selection.

Boundary	What it controls
Data boundary	Authoritative sources, derived data, retention, residency and access ownership.
Context boundary	What information is assembled for a model call, why it is included, provenance and tenant scope.
Knowledge boundary	Persistent enterprise knowledge vs transient workflow state and long-term memory.
State boundary	Durable/recoverable workflow state without making agent reasoning the system of record.
Execution boundary	Model proposals vs deterministic tool invocation, business transitions and external side effects.
Trust boundary	Identity, authorization, privacy, policy, evidence and approval outside model authority.
Failure boundary	Dependency degradation, retries, timeouts, duplicate delivery, partial completion and regional failure.
Economic boundary	Tokens, inference, retrieval and execution cost by tenant, workload and business capability.

Stage	Architectural question	Output	Production consequence
Business intent	What outcome must change, and what is the unacceptable business failure?	Outcome contract	Avoid optimizing the wrong objective
Requirement discovery	What must happen, who owns it, and what evidence is required?	Capability / interaction model	Defines boundaries
NFR analysis	What are latency, throughput, availability, consistency, privacy and recovery targets?	Measurable NFR contract	Drives capacity and execution model
Constraint analysis	What cannot move, fail, leak or become probabilistic?	Constraint-to-resolution map	Eliminates non-compliant designs
AI suitability	Where is probabilistic inference justified?	AI boundary	Prevents model output becoming system authority
Architecture options	Which design survives constraints with acceptable complexity/economics?	Selected target state	Makes trade-offs explicit
Detailed design	How do state, retries, identity, policy, data and dependencies behave?	Component contracts	Makes failure behavior testable
Validation	How do we prove workflow, AI quality, security, resilience and economics?	Evidence pack	Prevents demo-only acceptance
Production feedback	What did traffic, failures, cost and quality reveal?	Architecture evolution backlog	Feeds telemetry back into architecture

FROM AMBIGUITY TO ARCHITECTURE

Converting enterprise asks into explicit architecture

Phrases such as “autonomous assistant”, “enterprise knowledge”, “multi-agent workflow” or “real-time AI” are not sufficient engineering requirements. The first architectural task is to force ambiguity into explicit dimensions: capability, actor, data, context, authority, latency, failure, compliance and economics.

Resolution chain

1. Separate the business outcome from the proposed technology.
2. Convert the outcome into functional capabilities and explicit system actors.
3. Translate quality expectations into measurable NFRs rather than adjectives.
4. Identify hard constraints: data locality, regulatory controls, tenant isolation, recovery objectives and cost ceilings.
5. Determine where probabilistic inference is justified and where deterministic control is mandatory.
6. Choose service boundaries around ownership, state, scaling and failure characteristics.
7. Choose execution semantics: synchronous, asynchronous, durable, resumable or human-gated.
8. Define model, knowledge, tool and agent interfaces independently so implementation choices remain replaceable.
9. Attach observability, evaluation and FinOps signals to the same correlation identity used by runtime execution.
10. Validate the architecture against production failure modes before optimizing for feature completeness.

Ambiguous statement	Question forced	Architectural resolution
"The agent should act autonomously."	Which actions can be model-proposed, policy-controlled or approval-gated?	Bounded autonomy; deterministic execution and approval gates.
"Use enterprise knowledge."	Which source is authoritative, tenant-scoped and evidence-bearing?	Controlled retrieval and provenance-aware context assembly.
"Support many customers."	What state can cross tenant boundaries?	Tenant identity propagated through control/data planes and tenant-scoped state.
"Use the best model."	Best for what quality threshold, latency class and cost budget?	Model abstraction and workload-aware routing.
"Be real time."	Which path is latency-sensitive and which work can be asynchronous?	Synchronous user path separated from asynchronous execution.
"Recover automatically."	What proves the last safe point and how are side effects made idempotent?	Durable checkpoints, action identity and deterministic recovery.

REQUIREMENT DISCOVERY

Turning business behavior into capability contracts and architectural ownership

Each requirement is mapped to an owning boundary, interaction contract, state semantics, failure semantics and evidence required for acceptance.

Functional contract patterns

- Every externally visible capability receives an explicit input contract, output contract and error taxonomy.
- Every long-running operation receives a correlation identifier, workflow identifier and tenant identifier.
- Every tool invocation receives an action identifier so retries can distinguish replay from a new side effect.
- Every model call carries workload metadata required for routing, evaluation, observability and cost attribution.
- Every retrieved context item carries provenance and authorization context before it reaches the model.
- Every policy decision is observable as a deterministic event rather than hidden inside prompt text.

Requirement class	Example	Acceptance evidence
Functional	Create and execute multi-step agent workflows	Workflow contract + end-to-end trace
Contextual	Retrieve tenant-authorized enterprise knowledge	Evidence with tenant + provenance metadata
Execution	Invoke enterprise tools through governed interfaces	Authorized tool trace + audit event
Interoperability	Delegate a task between agents	Capability contract + delegated-task trace
State	Resume a failed long-running workflow	Checkpoint + replay/compensation evidence
Security	Prevent cross-tenant data/execution leakage	Negative isolation tests + authorization traces
Quality	Detect retrieval/model regression	Evaluation run linked to release telemetry
Economic	Attribute inference cost to workload/tenant	Token/model usage linked to task

Dependency	Failure mode	Required runtime behavior
Model provider	Timeout, rate limit, degraded quality	Bounded retry; route to allowed alternative; preserve task state
Vector retrieval	Latency spike/unavailable partition	Bound retrieval deadline; approved fallback or fail closed where required
Knowledge graph	Relationship traversal unavailable	Do not silently replace relationship evidence when relationship semantics are required
Tool service	Duplicate/partial side effect	Action identity, idempotency and result reconciliation
Event backbone	Duplicate delivery	Consumer idempotency keyed by event identity and business action

FUNCTIONAL REQUIREMENTS & BOUNDARY OWNERSHIP

A platform fails architecturally when multiple capabilities own the same state or policy. If each agent decides tenant authorization, isolation cannot be proven consistently. If each application embeds model selection, economics and provider failover become fragmented. If the agent owns durable business state, replay becomes coupled to probabilistic reasoning.

- Business systems remain authoritative for business state.
- The agent runtime owns execution state, not the enterprise system of record.
- The knowledge layer owns retrieval and evidence assembly, not business authorization.
- The model gateway owns model selection and provider abstraction, not tool permission.
- The MCP gateway owns tool authority, identity propagation and audit.
- The evaluation plane measures behavior; it does not become a runtime authorization mechanism.
- The observability plane records execution evidence without becoming a hidden source of business truth.

Capability	Owning boundary	Implementation mechanism	Proof
Tenant lifecycle	Control plane	Tenant configuration, identity propagation, policy configuration	Isolation/configuration tests
Agent registration	Control plane	Metadata, capabilities, versions, policy bindings	Registration contract + policy trace
Durable workflow	Agent runtime	LangGraph state graph, checkpointing, controlled transitions	Resume-after-failure test
Knowledge retrieval	Knowledge service	GraphRAG + vector retrieval + context assembly	Evidence lineage + retrieval evaluation
Tool execution	MCP gateway	Discovery, authorization, policy checks, audit	Allowed/denied execution traces
Agent delegation	A2A gateway	Capability contract + delegated task identity	Delegation trace with authority scope
Model inference	Model gateway	Provider abstraction, routing, fallback, semantic cache	Latency/quality/cost telemetry
Safety controls	Governance plane	Presidio, policy enforcement, evaluation gates	Policy decision + adversarial evidence
Operational telemetry	Observability plane	OpenTelemetry across runtime/inference/dependencies	Trace-to-metric/evaluation correlation

NFR ARCHITECTURE

Latency, scale, availability, consistency, security, recovery and economics

The useful unit is not “model latency”; it is end-to-end task latency. A representative budget is decomposed into gateway admission, authentication/policy, retrieval, context assembly, model queueing, inference, tool execution and response serialization. Each segment receives a deadline.

NFR	Architecture implication	Mechanism	Reference evidence
P95 inference latency	Separate hot path from asynchronous work	Semantic caching, context minimization, model routing, bounded retrieval	3.4 s → 870 ms
Concurrent sessions	Avoid process-local workflow state	Durable state + horizontally scalable runtime on AKS	1,000+ sessions in detailed evidence set
Availability	Remove single control dependency from runtime path	Failure isolation, timeouts, graceful degradation, recovery controls	99.94% in detailed evidence set
Tenant isolation	Authorization cannot depend on prompt behavior	Tenant-scoped identity, policy and state boundaries	Cross-tenant negative tests
AI quality	Measure output independently of application success	RAGAS, DeepEval, Giskard, PyRIT workflows	Release/regression evidence
Cost	Model choice becomes an economic policy	Routing, semantic cache, usage telemetry	38% lower LLM serving cost
Operational recovery	Runtime preserves enough state to resume	Checkpointing, action identity, recovery orchestration	90% fewer downtime incidents stated in source

Segment	Control	Failure response
Admission + policy	Hard timeout	Reject/defer before expensive inference
Retrieval	Bounded parallel retrieval	Required evidence or approved degraded path
Context assembly	Token budget	Drop lower-value context by deterministic rules
Inference	Routing + provider timeout	Approved model class or bounded fallback
Tool execution	Per-tool timeout + action identity	Retry only when idempotency permits
Overall task	Deadline propagation	Cancel downstream work after business deadline

CONSTRAINTS & CONSTRAINT RESOLUTION

Constraint resolution pattern

1. Name the constraint in operational terms.
2. Identify the architectural boundary that can enforce it deterministically.
3. Move enforcement as close as possible to the authority being protected.
4. Record the additional latency, operational burden and failure mode introduced by the control.
5. Define the evidence that proves the constraint remains satisfied after deployment.

Constraint	Why it matters	Resolution mechanism	Trade-off accepted
Tenant isolation	Shared runtime can leak state/context if identity is implicit	Tenant identity and policy propagated through control/data planes; tenant-scoped state/retrieval	More policy metadata/enforcement points
Probabilistic behavior	Models can produce invalid/unauthorized actions	Model proposes; deterministic policy, authorization and workflow controls decide	More orchestration/policy latency
Long-running workflows	Process-local state is lost on failure	Durable workflow state + checkpoints	Persistent state/replay complexity
Provider variability	Quality, latency and limits change independently	Model abstraction + workload-aware routing	Routing policy/evaluation burden
Enterprise knowledge	Vector similarity does not preserve relationships/authority	GraphRAG + vector retrieval + explicit context assembly	Dual retrieval/indexing complexity
Tool side effects	Retries can duplicate business actions	MCP gateway + action identity + idempotent contracts	Tool contract changes/audit storage
Operational visibility	Agent traces alone do not prove business outcome	OpenTelemetry correlation across state/inference/tools/policy/evaluation	Higher telemetry volume
Inference economics	Frontier routing for every request is expensive	Semantic cache + workload-aware routing	Routing/cache correctness concerns

PRINCIPAL-LEVEL DECISION TEST

If a constraint can only be satisfied by asking the model to behave correctly, it is not yet an architecture control. Move the requirement into identity, policy, workflow state, tool authorization, data access or deterministic validation.

ARCHITECTURE OPTIONS & DECISION LOGIC

Selection criteria

- Architectural integrity: clear ownership of state, policy and authority.
- Failure behavior: bounded blast radius and deterministic recovery.
- Security: enforceable tenant and tool boundaries.
- Performance: predictable hot-path latency and scalable asynchronous work.
- Economics: ability to route, cache and attribute model usage.
- Evolution: provider/framework replacement without rewriting business capabilities.
- Operational proof: telemetry and evaluation hooks at the platform boundary.

Decision	Option considered	Why rejected	Selected direction
Runtime model	Independent agent applications	Duplicates state, policy, tool access and observability	Reusable platform runtime with explicit control/data planes
Workflow execution	Pure synchronous request chain	Long-running work and dependency failure become request failures	Durable orchestration + asynchronous execution where appropriate
Knowledge	Vector-only retrieval	Relationship-heavy enterprise queries need graph semantics	GraphRAG + vector retrieval
Model integration	Provider-specific SDK per application	Creates coupling and prevents platform-wide routing/economics	Model gateway + capability abstraction
Tool governance	Agent directly calls tools	Authorization becomes distributed and difficult to audit	MCP gateway with identity, policy and audit
Agent delegation	Direct peer-to-peer calls	Authority scope and audit become ambiguous	A2A through explicit capability/identity contracts
State	Agent memory as system of record	Couples model execution to business truth	Separate workflow state, session context, long-term memory and business state
Caching	Cache every response	Semantic equivalence unsafe when context/authorization differs	Context-aware semantic caching with policy boundaries

TRADE-OFF LEDGER

Intentional complexity, accepted rationale and payoff

The platform deliberately treats inference cost as an architecture variable. Routing to a higher-capability model may improve quality but can increase latency and serving cost. Semantic caching reduces repeated inference only when context freshness, tenant scope and authorization make reuse safe. The target is not “cheapest model”; it is the lowest cost that satisfies the workload quality, latency and policy contract.

Trade-off	Complexity introduced	Why accepted	Payoff
Control plane + data plane	Two architectural paths + synchronization concerns	Policy/tenant control should not be entangled with runtime execution	Stronger isolation + reusable governance
Durable state	Checkpoint storage, versioning, recovery logic	Long-running work cannot depend on process lifetime	Resumability + controlled recovery
Graph + vector retrieval	Two indexing/retrieval paths	Relationships and semantic similarity solve different problems	Relationship-aware context + retrieval precision
Model gateway	Routing policy, adapters, fallback	Applications should not own provider economics/failure handling	Provider portability + cost control
MCP governance	Gateway hop, authorization, audit	Tool execution is a trust boundary	Controlled side effects + auditable authority
Evaluation plane	Offline datasets, regression jobs, production signals	Functional tests cannot measure model/retrieval behavior	Detectable quality drift
OpenTelemetry across AI	High-cardinality telemetry/storage cost	Model calls, tools and state transitions need common causal trace	Diagnosis + cost-quality correlation

870 ms P95 AI response latency	38% Lower LLM serving cost	99.94% Availability in detailed evidence set
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COMPONENT CATALOG

Technical responsibility, state, scaling and failure characteristics

Interface discipline

- Runtime-to-model calls carry workload class, tenant identity, context metadata and correlation identity.
- Runtime-to-tool calls carry action identity, authorization context and deadline.
- Knowledge responses carry evidence identifiers and provenance, not only text snippets.
- Evaluation results reference model, prompt, retrieval configuration and workflow version so regressions can be attributed.
- Telemetry spans preserve causal linkage from user intent through state transitions, retrieval, model inference, tool execution and final outcome.

Component	Primary responsibility	State	Scaling model	Critical failure concern
Tenant/policy control plane	Tenant configuration, policy, platform metadata	Persistent control state	Horizontal services + durable stores	Incorrect policy propagation
Agent runtime	Workflow execution/state transitions	Durable workflow state	Horizontal execution workers	Lost/duplicated transitions
Context engine	Context assembly from knowledge, memory, request	Transient context + references	Horizontal, latency-sensitive	Oversized/unauthorized context
Knowledge service	Graph/vector retrieval	Persistent indexes	Partitioned retrieval services	Stale/incomplete evidence
MCP gateway	Tool identity, policy, authorization, audit	Audit/invocation metadata	Horizontal gateway tier	Unauthorized/duplicate side effect
A2A gateway	Capability discovery/delegated tasks	Delegation metadata	Horizontal protocol tier	Authority escalation
Model gateway	Provider abstraction, routing, fallback	Routing config + telemetry	Horizontal gateway + provider capacity	Provider outage/bad routing
Semantic cache	Reuse semantically equivalent inference	Cache entries + validity context	Distributed cache	Stale/cross-tenant data
Evaluation plane	Quality, regression, adversarial testing	Datasets/results	Batch/scheduled workloads	Unmeasured drift
Observability plane	Traces, metrics, logs, cost signals	Telemetry data	Scalable telemetry pipeline	Signal loss during incidents

COMPONENT BOUNDARIES & AUTHORITY MODEL

Where authority lives and where probabilistic behavior stops

MCP execution chain

1. Model produces a structured tool intent.
2. Runtime validates that the requested tool is registered for the current agent and tenant.
3. MCP gateway resolves tool identity and capability metadata.
4. Identity and tenant context are propagated to the authorization layer.
5. Policy checks evaluate data sensitivity, operation type, tenant scope and approval requirements.
6. If authorized, the gateway issues an action identity and invokes the tool.
7. The result is returned to the runtime as data; the business system remains authoritative for the side effect.
8. The invocation, policy decision, result and latency are recorded as correlated audit and telemetry events.

A2A delegation boundary

Agent-to-agent delegation is treated as capability delegation rather than unrestricted peer trust. The receiving agent receives an explicit task contract and bounded authority. The originating agent remains accountable for the delegated workflow, while the platform records the delegation chain so the final business outcome can be traced through each agent boundary.

Layer	Can propose	Can authorize	Can execute side effects
Foundation model	Reasoning, classification, tool intent	No	No
Agent runtime	Workflow transition according to defined graph	Within runtime policy	Only through governed interfaces
Policy engine	Policy decision	Yes, for defined policy domain	No
MCP gateway	No business decision	Yes, tool authorization	Yes, through controlled contract
Business service	Business validation	Yes, for business state	Yes
Human approver	Decision	Yes, where required	Yes through authorized workflow

BOUNDARY RULE

The model may select among capabilities the platform exposes. It does not create new authority by generating a tool name, policy statement or delegation request.

STATE, CHECKPOINTS & RECOVERY

Durable execution, idempotency and deterministic recovery

Checkpoint semantics

A checkpoint must represent more than a serialized conversation. It captures workflow version, current node, validated inputs, relevant references, completed action identities, pending work and recovery metadata. This lets the runtime distinguish “model has proposed an action” from “external side effect has committed.”

Failure and recovery sequence

1. Detect dependency failure or worker loss.
2. Stop advancement from an unverified state.
3. Load the last valid checkpoint.
4. Determine which actions were committed, pending or unknown using action identity and external result reconciliation.
5. Replay only deterministic transitions that are safe to replay.
6. Retry idempotent actions within policy limits.
7. Escalate ambiguous side effects rather than duplicating them.
8. Persist the new checkpoint before advancing the workflow.
9. Emit a recovery trace linked to the original task and incident context.

Why simple retries are insufficient

A retry after a timeout cannot prove whether the remote service committed the first request. In agentic systems this is particularly dangerous because the model may generate a new plan after the timeout. Recovery therefore operates on execution identity and durable state, not on conversation history.

State type	Owner	Lifetime	Recovery rule
Authoritative business state	Business system	Business-defined	Recover through business transaction semantics
Workflow state	Agent runtime	Task lifetime	Resume from validated checkpoint
Session context	Context layer	Session	Reconstruct or expire according to policy
Long-term memory	Memory service	Policy-defined	Versioned retrieval; never authoritative business truth
Model output	Execution trace	Audit-defined	Immutable evidence; not authoritative state
Tool action state	Tool/business service	Operation lifetime	Idempotent action identity + reconciliation

SECURITY & GOVERNANCE

Identity, authorization, privacy, policy enforcement, evidence and adversarial evaluation

Security is enforced at execution boundaries rather than delegated to the model. Identity, tenant isolation, authorization, secrets, data handling and tool permissions are deterministic platform responsibilities.

Governance enforcement locations

- Identity is established before model and tool access, not inferred from prompt content.
- Tenant scope is carried as machine-readable execution context.
- Authorization is checked at the resource boundary that owns the protected capability.
- Privacy controls operate before sensitive data is sent to downstream inference or tools.
- Policy decisions are versioned and observable so post-incident reconstruction is possible.
- Human approval is used where business risk cannot be bounded sufficiently by deterministic policy.

Threat	Weak approach	Architectural control
Cross-tenant retrieval	Prompt instruction to use tenant data only	Tenant identity enforced at retrieval and policy boundaries
Prompt injection	Prompt sanitization only	Input handling, retrieval controls, tool authorization and adversarial testing
Tool misuse	Agent prompt says "use tools responsibly"	MCP gateway, capability registration, authorization and policy evaluation
Sensitive data leakage	Application developer discipline	Presidio + policy enforcement at platform boundaries
Unauthorized delegation	Peer agents trust each other	A2A identity + bounded capability delegation
Model drift	Manual review after incidents	Evaluation gates + production quality telemetry
Evidence tampering	Mutable logs	Immutable provenance + controlled evidence storage

VALIDATION & RELEASE GATES

Proving architecture, workflow, AI quality, resilience, security and economics

Release gates

- Schema and API contract tests pass.
- Security and tenant-isolation negative tests pass.
- Retrieval evaluation remains within the approved quality band.
- Model and agent evaluation does not regress beyond the release threshold.
- Performance test demonstrates target latency under the defined traffic profile.
- Failure-injection tests demonstrate bounded recovery behavior.
- Observability confirms every critical execution path has a correlated trace.
- Cost telemetry confirms workload attribution and routing policy behavior.

Validation layer	Question	Evidence
Contract	Do service/tool interfaces obey contracts?	Schema and contract tests
Workflow	Can runtime complete and resume correctly?	State-machine + failure-injection tests
Retrieval	Is the right evidence retrieved with acceptable precision?	RAGAS + retrieval benchmark
Model	Does selected model meet quality contract?	Evaluation dataset + regression results
Agent trajectory	Does agent take bounded/useful steps?	Trajectory evaluation + tool-use traces
Security	Can unauthorized/injected actions cross boundaries?	Negative tests + PyRIT/Giskard scenarios
Performance	Does system meet latency/concurrency targets?	Load test + dependency-level breakdown
Resilience	What happens when dependencies fail?	Fault injection + recovery evidence
Economics	What does a successful task cost?	Token/inference/task-cost telemetry

EVIDENCE PRINCIPLE

A green functional test is not production proof for an agentic platform. Production readiness requires evidence across deterministic workflow behavior, probabilistic quality, trust boundaries, failure recovery, operational telemetry and unit economics.

DEPLOYMENT & PRODUCTION ROLLOUT

Cloud-native deployment with separation of control, execution, evaluation and operations

Deployment safety

- Version workflow definitions independently from long-running workflow instances.
- Version policy and evaluation configurations so production traces identify the rules used at execution time.
- Do not allow a new model or retrieval configuration into production without evaluation evidence.
- Keep rollback paths deterministic; model rollback should not require replaying an already committed business side effect.
- Use health signals from both infrastructure and AI behavior before increasing traffic.

Production rollout sequence

1. Validate infrastructure and dependency health.
2. Deploy immutable service versions.
3. Run contract and smoke tests.
4. Run evaluation and regression gates.
5. Enable controlled traffic.
6. Observe latency, quality, tool failures and cost.
7. Increase traffic only when composite production signals remain within policy.
8. Retain release metadata for incident reconstruction.

Deployment concern	Design choice	Reason
Runtime	AKS for agent execution and platform services	Independent scaling and failure isolation
Control services	Separate deployment lifecycle from runtime workloads	Policy/config changes should not restart active workflows
Inference	Provider abstraction through model gateway	Routing/provider evolution without application coupling
Knowledge	Dedicated graph + vector retrieval services	Independent scaling and operational ownership
Observability	OpenTelemetry across services	One causal trace across distributed execution
Evaluation	Separate evaluation workloads from user traffic	Prevent tests competing with production inference
Infrastructure	Infrastructure-as-code + controlled CI/CD	Repeatability and auditable deployment

PRODUCTION READINESS & INCIDENT RESPONSE

Operational SLOs, observability, degradation and recovery

A useful incident trace follows: business intent → tenant identity → workflow version → state transitions → retrieved evidence → model calls → policy decisions → tool actions → external result → final business outcome → cost. This is materially more useful than a model prompt log because it exposes deterministic and probabilistic parts separately.

Signal	Why it matters	Operational action
Inference P95	User experience + capacity	Scale, reroute or reduce context budget
Retrieval latency	Consumes inference budget	Tune index, parallelize or bounded fallback
Tool latency/failure	Determines action completion	Trip dependency controls; retry only when safe
Agent completion rate	Measures workflow success	Inspect state transitions, tool failures, model behavior
Evaluation pass rate	Detects behavioral regression	Block release or alter routing
Cost per successful task	Connects AI usage to economics	Change routing, caching or context policy
Tenant error concentration	Detects isolation/workload-specific problems	Throttle or isolate affected workload

Failure	Immediate control	Degradation	Recovery
Model provider unavailable	Timeout + circuit control	Route to allowed provider or defer	Restore provider + drain queued work
Retrieval unavailable	Bounded timeout	Approved deterministic fallback where valid	Rebuild/restore retrieval
Tool unavailable	Action timeout	Pause workflow or request approval	Resume from checkpoint
Worker loss	Lease expiration	No new state advancement	Resume from checkpoint
Telemetry degradation	Local buffering where supported	Protect critical audit signals	Backfill/reconcile telemetry
Regional failure	Traffic/dependency health detection	Fail over per data/service policy	Reconcile state + restore capacity

TRACEABILITY

Business requirement → architectural decision → implementation mechanism → validation evidence → outcome

Every major architecture claim should be explainable in five steps: what problem existed, what constraint made the obvious solution insufficient, what decision changed the architecture, what mechanism implemented the decision, and what production evidence showed that the decision worked.

Architecture decision logic

1. Start with the constraint, not the technology.
2. Explain the boundary you introduced and why ownership belongs there.
3. State the alternative you rejected and the failure mode it created.
4. Describe runtime semantics: state, retry, timeout, authorization and side-effect behavior.
5. Close with the measurable production consequence.

Business need	Architectural decision	Implementation mechanism	Evidence / outcome
Operate multiple tenants safely	Tenant scope as platform boundary	Control/data plane separation + tenant-scoped state/policy	Governed isolation + concurrency evidence
Allow autonomy without uncontrolled actions	Separate model intent from deterministic execution	Agent runtime + MCP gateway + policy controls	Controlled tool execution + auditable authority
Use enterprise knowledge with relationships	Combine graph and semantic retrieval	Neo4j GraphRAG + vector retrieval + context assembly	Retrieval evidence + provenance
Avoid provider lock-in	Abstract model capabilities from providers	Model gateway + routing across approved models	Provider flexibility + workload-aware inference
Reduce redundant inference	Cache at semantic context boundary	Redis + GPTCache	38% lower LLM serving cost
Reduce user-visible latency	Optimize entire inference path	Routing + caching + context control + dependency boundaries	3.4 s → 870 ms
Recover long-running workflows	Make execution state durable	Checkpointed runtime + idempotent action semantics	Reduced downtime incidents
Make AI quality measurable	Evaluation as platform capability	RAGAS, DeepEval, Giskard, PyRIT	Release/regression evidence
Operate reliably	Instrument complete causal path	OpenTelemetry across runtime/inference/dependencies	99.94% availability in detailed evidence set

DEEP FLAGSHIP ARCHITECTURE NARRATIVE

CTO / VP AI / Principal Architect review

Platform mandate

Enterprise multi-tenant Agentic AI platform and Enterprise AI Operating System for enterprise SaaS workloads. The architectural problem was not simply to build an agent. It was to create reusable platform primitives so multiple AI capabilities could operate without each application independently solving tenant isolation, orchestration, knowledge access, model integration, tool authority, evaluation and operations.

Agent runtime mechanics

LangGraph was used as the execution substrate for controlled multi-step workflows. The architectural distinction is between a graph transition and a model decision: the model can influence the next action, but the runtime validates whether that action is legal within the workflow, tenant and policy context. Durable execution semantics keep workflow state separate from business truth.

Knowledge and context

GraphRAG was used where relationship structure mattered, while vector retrieval supported semantic matching. The context layer selects evidence, controls payload size and maintains provenance. This avoids treating a vector store as the enterprise knowledge authority and avoids pushing unrestricted retrieval output directly into inference.

MCP and A2A

MCP provides a governed boundary for tool discovery and execution. The gateway prevents the model from directly possessing enterprise credentials or independently determining authorization. A2A applies the same principle to agent delegation: capabilities are delegated through explicit contracts rather than implicit peer trust.

Inference economics

The model layer separates application capability from individual providers. Routing considers workload requirements such as quality, latency and serving economics. Redis and GPTCache reduce repeated inference when semantic equivalence and authorization context make reuse safe. This is a FinOps mechanism, not merely a performance optimization.

Quality and adversarial evaluation

RAGAS and DeepEval provide measurable evaluation of retrieval and model behavior; Giskard and PyRIT support adversarial testing. The platform-level design makes evaluation a repeatable release concern rather than a one-time application test.

EVALUATION & QUALITY ENGINEERING

Evaluation as a release-control surface

Evaluation is a release-control mechanism, not a dashboard attached after deployment. The platform separates retrieval quality, response quality, tool-use correctness, policy adherence, latency and cost so a change can be accepted or rejected against explicit contracts.

How evaluation closes the architecture loop

- Every evaluation result is tied to a workflow, model configuration, retrieval configuration or policy version so failures can be attributed to a change.
- Golden datasets are complemented by adversarial and boundary-condition cases; passing average quality is insufficient when a small failure class can create an enterprise control breach.
- Evaluation artifacts are versioned with the release candidate and retained as evidence for production-readiness review.
- Production observations feed back into the evaluation corpus: recurring failures become regression cases rather than tribal knowledge.

Layer	What is evaluated	Mechanism	Decision produced
Retrieval	Relevance, context precision/recall, provenance	RAGAS-style retrieval evaluation + curated evidence sets	Accept, tune, change chunking or reject
Generation	Groundedness, relevance, instruction adherence	DeepEval-style task/model evaluation	Release gate or regression alert
Workflow	State transitions, tool selection, delegation, termination	Deterministic scenario suites against LangGraph workflows	Workflow contract pass/failure
Tool execution	Argument validity, authorization, side effects	MCP contract tests + negative authorization cases	Permit, deny or defect
Safety	Prompt injection, data leakage, unsafe tool use	Giskard + PyRIT adversarial suites	Block release or remediation
Performance	P95 latency, throughput, queue depth, saturation	Load/soak tests + production telemetry	Capacity/routing decision
Economics	Tokens, cache hit rate, model mix, cost/task	Usage metering by tenant/workflow	Budget/routing policy change

Decision	Rejected direction	Why	Chosen resolution
Quality scoring	One composite AI score	Hides retrieval, generation, policy or infrastructure failure	Separate quality dimensions + explicit gates
Evaluation coverage	Offline benchmark only	Misses drift, tool failures and production context	Offline suites + production sampling/telemetry
Judging	LLM judge for everything	Judge variance/circularity can hide deterministic defects	Deterministic assertions + curated references + model-based evaluation

SECURITY ARCHITECTURE & RUNTIME CONTROLS

Zero-trust execution boundary

Zero-trust execution sequence

1. Authenticate the caller and bind a tenant-scoped execution context.
2. Resolve policy before tool or data access; do not allow the model to bypass the policy path.
3. Classify the requested operation by risk, required data, side-effect scope and approval requirement.
4. Authorize the exact capability and resource; issue least-privilege execution credentials; then execute.
5. Emit an audit event containing decision context, policy version, principal, tenant, tool, resource and outcome.
6. On policy uncertainty, identity failure or missing authorization evidence, fail closed rather than falling back to model judgment.

Security-specific AI controls

- Presidio-style detection and redaction is applied before sensitive content is allowed into model context where policy requires it.
- Prompt-injection resistance is layered: retrieval isolation, instruction hierarchy, tool authorization and adversarial testing — not a single prompt.
- Secrets never become model context; enterprise credentials remain in controlled execution infrastructure.
- Human approval is reserved for high-risk or irreversible actions, preserving automation for lower-risk operations.

Boundary	Control	Authoritative decision	Failure behavior
Tenant	Tenant identity propagated through request/workflow state	Policy engine + service authorization	Deny cross-tenant access
Identity	User/workload identity + short-lived credentials	Identity provider + gateway	Fail closed
Data	Classification, redaction, minimization	Data policy before context assembly	Suppress/transform sensitive content
Tool	MCP registry, capability metadata, risk class	MCP gateway policy	Deny or require approval
Delegation	A2A contract + delegated capability scope	Delegation policy	Reject unauthorized delegation
Model	Approved model/provider policy	Model gateway	Route to approved model or fail
Runtime	Network/workload isolation	Kubernetes/cloud policy controls	Quarantine/restart
Audit	Immutable event record for security-relevant actions	Audit plane	Alert on missing evidence

OBSERVABILITY & TELEMETRY ARCHITECTURE

Correlating distributed-system behavior with AI behavior

AI observability must explain both distributed-system behavior and probabilistic behavior. The platform correlates request, workflow, retrieval, model, tool, policy, evaluation and infrastructure telemetry under a common trace identity without treating prompts or sensitive content as automatically safe to log.

Tracing design

- OpenTelemetry provides the distributed trace backbone; spans represent architectural boundaries rather than every internal function call.
- Long-running agent workflows retain stable workflow identity across asynchronous queue transitions and retries.
- Sensitive prompts and retrieved documents are not indiscriminately copied into telemetry; references, classifications, hashes or redacted fields are used where evidence is required.
- Model gateway spans record model selection reason, latency, token usage and outcome classification, allowing routing decisions to be audited without exposing unnecessary content.

Signal	Examples	Correlation key	Operational use
Traces	Request → workflow → retrieval → model → MCP → downstream	Trace ID, workflow ID, tenant ID	Latency attribution + failure localization
Metrics	P50/P95/P99, throughput, queue depth, cache hit, token rate	Tenant, workflow, model, region	Capacity, SLO, FinOps
Logs	Policy decision, tool outcome, deployment event, model error	Trace ID + event ID	Forensics + incident response
AI quality	Groundedness, retrieval relevance, refusal rate, tool correctness	Workflow/evaluation version	Quality regression detection
Security	Denied calls, policy violations, injection detections	Principal, tenant, policy version	Threat detection
Economics	Input/output tokens, model mix, cache reuse, cost estimate	Tenant, task, model	Budget + routing control

Observed symptom	Likely boundary	Diagnostic evidence
Latency spike	Queue, retrieval, model or tool dependency	Span distribution + queue age
Quality regression	Retrieval, context assembly or model route	Evaluation metrics correlated to configuration version
Cost increase	Routing, cache miss or context growth	Token telemetry + model-mix distribution
Tool failures	MCP gateway or downstream service	Tool result, policy decision + downstream latency
Intermittent workflow failure	Checkpoint or dependency	State transition, retry count + idempotency key

RESPONSIBLE / EXPLAINABLE / ETHICAL AI

System property rather than a model claim

Responsible, explainable and ethical AI is implemented as a system property. The platform does not claim that a language model is inherently explainable or ethical. Instead, it makes data provenance, decision boundaries, human authority, policy enforcement, evaluation and appeal paths explicit.

Explainability boundary

The platform does not expose hidden chain-of-thought as an explanation mechanism. It records decision-relevant evidence: retrieved sources, policy checks, tool actions, confidence or quality signals where available, and the deterministic outcome. Where a workflow produces a consequential recommendation, the system distinguishes model-generated rationale from authoritative business rules and source evidence.

Ethical-AI operating loop

1. Define intended use and prohibited use before model integration.
2. Identify affected data subjects, sensitive attributes and potential harm modes.
3. Create task-specific evaluation cases rather than relying on generic model benchmarks.
4. Enforce data and action boundaries at runtime.
5. Monitor production outcomes for drift, unexpected behavior and disparity signals.
6. Route material incidents into remediation, evaluation regression suites and governance review.

Concern	System control	Evidence
Transparency	Record model, prompt policy, retrieval sources and decision path at appropriate evidence level	Trace + audit records
Explainability	Expose retrieved evidence, rule outcomes and decision factors where workflow permits	Evidence bundle + user-facing rationale
Human oversight	Approval gates for high-risk or irreversible actions	Approval event with actor + timestamp
Fairness	Task-specific cohorts + disparity testing where required	Evaluation reports
Privacy	Data minimization, redaction and tenant isolation	Policy decisions + data-classification events
Safety	Adversarial testing + runtime policy enforcement	Red-team findings + blocked actions
Accountability	Named owner for model, workflow, policy and data domain	Ownership metadata + change history

TRADE-OFF

More controls add friction and latency, especially when human approval is required. The architectural resolution is risk-tiered control: low-risk informational actions remain automated, while high-impact or irreversible actions cross explicit approval and audit boundaries.

REAL-TIME MONITORING & SRE CONTROL LOOP

Observe → classify → decide → mitigate → verify

Real-time monitoring is treated as a closed control loop, not merely alerting. Telemetry is used to change routing, shed load, protect dependencies, trigger recovery and surface AI-quality or security regressions.

SRE mechanisms

- OpenTelemetry metrics and traces provide raw signal; Prometheus-style time-series monitoring supports alert evaluation and capacity analysis.
- Admission control prevents overloaded inference/tool dependencies from turning local saturation into platform-wide failure.
- Circuit breakers isolate unstable downstream dependencies; retries are bounded and only used when operations are safe to repeat.
- Autoscaling responds to concurrency and queue depth rather than CPU alone because inference/agent workloads can be accelerator- or token-bound.
- Recovery events are verified through post-action telemetry; failover is not considered successful merely because traffic moved.

Signal	Threshold class	Automated response	Human escalation
P95 inference latency	SLO breach / sustained degradation	Route to healthy model, reduce context, scale serving	Escalate persistent breach
Queue age	Backlog growth	Scale consumers, admission control, prioritize critical workloads	Escalate saturation
Model error rate	Provider degradation	Shift to approved alternate model	Provider incident
Cache hit rate	Unexpected drop	Inspect equivalence, TTL, authorization partitioning	Escalate persistent cost impact
Policy denial spike	Security anomaly	Rate-limit or quarantine workload	Security review
AI quality regression	Evaluation/production sample threshold	Hold rollout, route to safer configuration	Model/product owner review
Tool failure rate	Dependency degradation	Circuit break + idempotent retry	Service owner escalation
Token cost/task	Budget threshold	Cheaper approved model or context cap	FinOps review

CLOSED-LOOP PROPERTY

A latency incident can change routing; a quality incident can stop a rollout; a cost anomaly can change model selection; a security anomaly can reduce capability exposure. Each automated action is policy-bounded and observable.

CLOUD INFRASTRUCTURE & PLATFORM ENGINEERING

Deterministic substrate for the AI platform

Cloud infrastructure is the deterministic substrate on which the AI platform executes. The objective is to isolate control-plane responsibilities from data-plane workload execution, scale independently, make infrastructure reproducible and keep tenant isolation explicit.

Infrastructure decisions

- Kubernetes provides workload isolation and independent scaling for agent, retrieval and gateway services; it is not treated as the source of business truth.
- Event-driven execution prevents synchronous request chains from coupling every component's latency and availability.
- Terraform-style infrastructure-as-code makes topology reproducible and makes environment drift visible.
- Control-plane services remain small and stateful where authority is required; data-plane workers remain horizontally scalable and disposable.
- Regional resilience is designed around recoverable state, idempotent actions and replay semantics rather than simply running a second cluster.

Layer	Responsibility	Representative implementation	Architectural reason
Control plane	Tenant, policy, registry, deployment, governance state	Managed cloud services + Kubernetes control components	Stable governance independent of workload spikes
Agent data plane	Workflow execution + checkpointed state	LangGraph on AKS-class Kubernetes	Independent scaling + failure isolation
Event backbone	Asynchronous coordination	Azure Service Bus	Decouple producers/consumers + absorb bursts
Knowledge plane	Graph + semantic retrieval	Neo4j + vector retrieval services	Separate relationship traversal from semantic similarity
Inference plane	Model gateway + serving integration	Azure AI Foundry + approved providers	Provider abstraction + routing control
Cache plane	Reusable context/inference results	Redis + GPTCache	Reduce repeated latency/inference cost
Telemetry plane	Traces, metrics, evaluation	OpenTelemetry + monitoring systems	Operational evidence across boundaries
Security plane	Identity, policy, redaction, threat testing	Policy enforcement, Presidio, PyRIT, Giskard	Deterministic trust boundary
Resilience plane	Regional recovery + controlled failover	Cross-region deployment + recovery automation	Availability without conflating failover with correctness

Trade-off	Decision
Managed services vs full self-hosting	Use managed capabilities where they reduce undifferentiated burden; retain explicit control where latency, portability or governance requires it.
Single region vs multi-region	Use cross-region recovery for critical platform services; accept complexity because availability depends on more than application replicas.
Synchronous vs asynchronous	Keep user-facing boundaries synchronous only where required; move long-running orchestration and tool work to asynchronous execution.

INFERENCE ARCHITECTURE & MODEL SERVING

Provider abstraction, routing, caching, batching and economics

The inference layer isolates application workflows from model providers and serving mechanics. The application asks for a capability under explicit quality, latency, context and cost requirements; the model gateway decides which approved route can satisfy those constraints.

Serving mechanics

- Azure AI Foundry provides the managed model and AI integration boundary for the platform pattern; the application remains insulated from provider-specific APIs.
- Where dedicated serving is required, NVIDIA Triton-style inference serving provides a controlled runtime for model endpoints with explicit batching, concurrency and accelerator utilization characteristics.
- Semantic caching with Redis and GPTCache is applied only when authorization context, tenant scope and semantic equivalence permit reuse. Cache correctness is a security property, not merely a performance feature.
- Routing can use approved models such as GPT-4 and Claude 3.5 according to workload quality, latency and economics; routing policy remains deterministic even though model output is probabilistic.

Latency and cost controls

- Reduce context before reducing model quality: retrieve only evidence needed for the task, cap payload growth and reuse semantically equivalent results when safe.
- Use asynchronous execution for work that does not require a synchronous response, preventing long tool chains from consuming request threads.
- Measure token usage and model selection at tenant and workflow granularity so cost can be attributed and controlled.
- Protect tail latency with provider health checks, bounded retries, concurrency limits and workload-aware routing.

Stage	Inputs	Decision
Workload classification	Task type, quality class, latency class	Select routing policy
Context assessment	Token estimate, retrieval payload, cache eligibility	Trim, compress or reuse context
Policy check	Tenant, data class, approved models	Allow only permitted model routes
Provider health	Latency, error rate, capacity	Exclude unhealthy routes
Economics	Token cost, cache state, tenant budget	Prefer cost-efficient route within quality guardrails
Model execution	Prompt, context, tool state	Generate response or structured action
Post-processing	Schema validation, policy checks, evaluation signals	Accept, retry, repair or reject

FAILURE SEMANTICS

A model timeout is not automatically retryable. The platform distinguishes generation failure from tool side effects and uses idempotency and workflow checkpoints before repeating operations. If no approved route satisfies policy and SLO constraints, the system fails explicitly rather than silently degrading into an ungoverned model.

AI RED TEAMING & ADVERSARIAL SECURITY

Threat model → attack scenario → detection → mitigation → regression → production monitoring

Red teaming tests whether the architecture remains safe when the model, retrieved content, user input or tool ecosystem behaves adversarially. The target is not to prove that a model is safe; it is to expose failure paths and verify that deterministic boundaries contain them.

Red-team operating loop

1. Map attack paths to concrete architecture boundaries.
2. Create reproducible attack cases and record expected containment behavior.
3. Run tests before release and after material changes to model, retrieval, tools or policy.
4. Turn confirmed failures into regression tests and, where required, runtime controls.
5. Retain evidence with the release and audit record so security claims remain testable.

Attack class	Target	Test mechanism	Required architectural control
Prompt injection	User or retrieved content	PyRIT-style suites + crafted corpus cases	Instruction isolation, retrieval boundaries, tool authorization
Data exfiltration	Context assembly + tool access	Canary secrets + cross-tenant probes	Tenant isolation, redaction, least privilege
Tool abuse	MCP capabilities	Malicious args + escalation attempts	Schema validation, risk policy, approval
Agent delegation abuse	A2A boundary	Unauthorized capability delegation	Explicit contracts + delegated scope
Model manipulation	Routing + system instructions	Adversarial prompts + conflicting instructions	Policy precedence outside model
RAG poisoning	Knowledge layer	Malicious/misleading documents	Source trust, provenance, evaluation, quarantine
Resource exhaustion	Inference + workflow runtime	High concurrency + pathological context tests	Admission control, quotas, bounded context, autoscaling
Output manipulation	Downstream deterministic systems	Malformed structured outputs	Schema validation + execution sandbox

PRINCIPAL-ARCHITECT PRINCIPLE

The most important red-team result is not that an attack was blocked by a prompt. It is that malicious model output or instruction cannot directly cross into an authoritative enterprise action without passing the same identity, policy, authorization and audit boundaries as any other request.

AUDITABILITY & EVIDENCE ARCHITECTURE

Reconstructing why an AI-driven action occurred

Auditability means being able to reconstruct why an AI-driven action occurred, under which identity and tenant, using which model and knowledge, under which policy, with which tool actions and approvals, and what the authoritative system ultimately recorded.

Audit event design

- Every security- or business-significant action receives a unique event identity and correlation identifiers.
- Events are append-oriented and tamper-evident in the audit plane; operational logs are not the authoritative audit record.
- Evidence captures policy and configuration versions so a later reviewer can distinguish current behavior from the behavior that actually executed.
- Sensitive payloads are minimized. Auditability does not justify storing raw regulated content when a reference, hash or redacted representation provides sufficient evidence.
- Audit records survive workflow retries and are linked by execution identity so duplicate attempts can be distinguished from duplicate business actions.

Evidence element	Captured at	Why it matters
Request identity	Ingress	Principal, tenant and request origin
Policy version	Authorization + execution	Control set governing the action
Workflow version	Agent runtime	Links behavior to deployed orchestration logic
Knowledge references	Context assembly	Provenance without indiscriminate content logging
Model route	Model gateway	Explains provider/routing choice
Tool request + result	MCP gateway	Reconstructs external actions/outcomes
Approval	Human-in-the-loop boundary	Proves human authority where required
Checkpoint + idempotency key	State layer	Explains retry/recovery behavior
Evaluation result	Evaluation plane	Links quality/safety evidence to release
Final authoritative outcome	Business system	Distinguishes AI proposal from system-of-record state

Dimension	Observability	Auditability
Purpose	Operate and diagnose the platform	Prove and reconstruct material actions
Retention	Operationally optimized	Governance/evidence requirements

END-TO-END AI PRODUCTION CONTROL PLANE

One integrated production control loop

The architecture connects requirements, constraints, runtime, knowledge, inference, action, evaluation, security, observability, audit, real-time monitoring and FinOps into one production control loop.

Stage	Architectural action	Control / evidence
1. Business intent	Clarify outcome, actors, authority and unacceptable failure	Requirement + NFR baseline
2. Architecture boundary	Separate data, context, state, execution, trust and cost responsibilities	Target-state HLD + boundary map
3. Agent execution	Run controlled workflows with durable state	LangGraph workflow + checkpoint identity
4. Knowledge	Retrieve graph and semantic evidence with provenance	Neo4j + vector retrieval + context policy
5. Inference	Classify workload and route to approved model	Model gateway + health + FinOps policy
6. Action	Discover/execute capabilities through governed interfaces	MCP, A2A, authorization, approval
7. Evaluation	Measure quality, workflow correctness and safety	RAGAS, DeepEval, regression suites
8. Security	Contain identity, data and capability risk	Policy, redaction, least privilege, adversarial tests
9. Observability	Correlate distributed and AI behavior	OpenTelemetry + operational metrics
10. Audit	Preserve accountable evidence	Audit event chain
11. Real-time control	Detect SLO, quality, security or cost deviation	Automated mitigation + escalation
12. Resilience	Recover state/dependencies without duplicating actions	Checkpoint, idempotency, recovery verification

EXECUTIVE
ARCHITECTURAL
INVARIANT

Probabilistic AI can propose, rank, summarize, retrieve and reason over enterprise context. Deterministic systems retain authority over identity, policy, state transitions, external side effects, evidence and recovery. That separation is what makes the platform governable at enterprise scale.

Architectural conclusion

The platform is not an agent with infrastructure around it. It is an enterprise AI operating system in which agentic execution is one controlled subsystem. The value of the architecture is the set of explicit boundaries and feedback mechanisms that keep quality, security, reliability and economics under engineering control.

BUSINESS IMPACT

Architecture decision logic

Measured platform outcomes

3.4 s → 870 ms P95 AI response latency	38% LLM serving cost reduction	99.94% Platform availability
1,000+ Concurrent AI sessions — detailed evidence set	90% Fewer operational downtime incidents	TENANT-SCOPED Runtime/data/memory/event/telemetry isolation

Platform scale and availability

8,500+ Concurrent AI sessions — cover	99.97% Platform availability — cover	Architecture experience — cover
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METRIC CONTEXT

The supplied reference contains multiple evidence sets. Do not present both 1,000+ and 8,500+ as the same validated number, or 99.94% and 99.97% as the same availability measurement, without first confirming the intended project baseline. The document intentionally preserves the source values rather than inventing a reconciliation.

Architecture decision logic

“The architectural challenge was not building an agent; it was creating a reusable execution and control substrate. I separated governance from runtime execution, made tenant context and authority explicit, introduced durable workflow state, isolated model providers behind a gateway, combined graph and vector retrieval with provenance-aware context assembly, and put MCP/A2A behind deterministic authorization boundaries. I then closed the loop with evaluation, OpenTelemetry-based observability, audit evidence, real-time controls and FinOps. The important part was not the technology inventory; it was proving that the system could scale, recover, remain governed and produce measurable outcomes.”

TECHNOLOGY / PLATFORM REFERENCE

Technologies stated in the supplied reference

Domain	Technologies / mechanisms stated
Cloud / platform	Azure, AKS, Kubernetes, Azure AI Foundry, Azure Service Bus
Workflow / agents	LangGraph, controlled graph workflows, durable checkpoints
Knowledge	Neo4j GraphRAG, vector retrieval, provenance, ACL filtering, context budgeting
Interoperability	MCP, A2A, capability contracts, delegated identity
Inference	GPT-4, Claude 3.5, provider abstraction, routing, fallback, semantic caching
Serving	NVIDIA Triton-style serving where dedicated serving is required; batching, concurrency, accelerator utilization
Cache	Redis, GPTCache, authorization-aware semantic cache
Messaging	Kafka / durable event coordination in core architecture; Azure Service Bus in cloud platform supplement
Security / privacy	Zero-trust, policy enforcement, Presidio, short-lived credentials, least privilege
Evaluation	RAGAS, DeepEval, Giskard, PyRIT, curated datasets, regression suites
Observability	OpenTelemetry, Prometheus-style monitoring, traces, metrics, logs
IaC / delivery	Terraform-style infrastructure-as-code, controlled CI/CD, immutable service versions
Reliability	Timeouts, circuit breakers, bounded retries, backpressure/admission control, idempotency, DLQ, recovery checkpoints
FinOps	Token metering, model mix, cache reuse, cost/task, tenant/workflow attribution, routing policy

ARCHITECTURE NOTE

Technology is treated as implementation evidence, not the architecture itself. The architecture is the ownership model, execution semantics, trust boundaries, failure behavior, economic controls and validation evidence around those technologies.

ARCHITECTURE STANDARD – CLOSING POSITION

Build the platform once. Make enterprise AI capabilities reusable, governed, observable and economically operable.

The architectural standard is not whether an agent can perform a task. It is whether the resulting system can explain:

- who had authority;
- what evidence was used;
- what decision was made;
- what action was executed;
- what failed;
- how it recovered;
- what it cost.

FINAL ARCHITECTURAL POSITION

Probabilistic AI belongs inside deterministic systems. The model can propose an action; the platform decides whether that action is permitted, how it is executed, whether it is committed, and what evidence is retained.

Architecture-first • Production-oriented • No client identity disclosed

Unified premium architecture case study • Diagrams intentionally excluded to eliminate truncation and right-edge rendering issues.